

Homework 4: Logistic Regression

DATA 627 - Statistical Machine Learning, Fall 2025

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1 What are the Odds?

a) What is the expected fraction, p , of people with an odds ratio of 0.10 of defaulting on their credit card payment who will in fact default?

- **Answer:** The odds of defaulting on credit card payment is 0.10. Therefore, the probability (or expected fraction) of defaulting is given by $p = \frac{odds}{1+odds}$. Hence:

$$p = \frac{odds}{1 + odds} = \frac{0.10}{1 + 0.10} = \frac{0.10}{1.10} = 0.09$$

So, approximately, 9% of people with odds of 0.10 will default on their credit card payment.

b) Suppose an individual has a 20% chance of defaulting on their credit card payment. What are the odds they will default?

- **Answer:** With $p = 0.20$, let's convert the probability to odds:

$$odds = \frac{p}{1 - p} = \frac{0.20}{1 - 0.20} = \frac{0.20}{0.80} = 0.25$$

Therefore, the odds of defaulting are 0.25, or 1 to 4.

2 What's it take to get an A?

Suppose we collect data on a group of students in a undergraduate statistics class using the following variables:

- X_1 : hours studied
- X_2 : undergrad GPA, and
- Y : receive an A (1) or Not an A (0).

We fit a logistic regression and produce estimated coefficients $\beta_0 = -3$, $\beta_1 = 0.06$, and $\beta_2 = 1.4$.

The fitted response function is:

$$\hat{\pi} = \frac{\exp(-3 + 0.06(Hours) + 1.4(GPA))}{1 + \exp(-3 + 0.06(Hours) + 1.4(GPA))}$$

a) Estimate the probability a student who studies for 15 hours and has an undergrad GPA of 3.0 gets an A in the class.

```
p = exp(-3 + (0.06*15) + (1.4*3)) / (1 + exp(-3 + (0.06*15) + (1.4*3)))
p
```

```
## [1] 0.8909032
```

- **Answer:** The estimated probability of getting an A is about 0.891 or 89.1%.

b) How many hours would the student in part (a) need to study to have a 95% (predicted) chance of getting an A in the class?

- **Answer:** Solving the equation above for X_1 (hours of study) when $p = 0.95$, gives the results below:

$$0.95 = \frac{\exp(-3 + 0.06(Hours) + 1.4(3))}{1 + \exp(-3 + 0.06(Hours) + 1.4(3))}$$

I found that $X_1 = 29.07$.

Therefore, the student would need to study approximately 29.07 hours to have a 95% predicted chance of getting an A.

3 Tuning the Titanic Model for Testing Data

The {DALEX} package's titanic data set includes a complete list of the passengers and crew along with additional variables to include survived.

- If you install the DALEX package (using the console), you can access it with `DALEX::titanic`.
- Then, remove NAs for age, fare, and embarked.

In class, we selected the threshold that minimized the training error classification rate. Use the reduced model with `age`, `gender`, and `class` as predictors of `survived`.

a) Split the data into training and testing data using 75% of the data for training. Set a seed of 1234 for the random number generator so the results are reproducible. Create a model and then determine which threshold minimizes the prediction error classification rate for the testing data.

1. Load the data and remove NAs for age

```
titanic <- DALEX::titanic
titanic <- tidyr::drop_na(titanic, c(age, fare, embarked))
glimpse(titanic)
```

```
## Rows: 2,179
## Columns: 9
## $ gender    <fct> male, male, male, female, female, male, male, female, male, m~
## $ age       <dbl> 42.0000000, 13.0000000, 16.0000000, 39.0000000, 16.0000000, 2~
## $ class     <fct> 3rd, 3rd, 3rd, 3rd, 3rd, 3rd, 2nd, 2nd, 3rd, 3rd, 3rd, 3rd, 3~
## $ embarked  <fct> Southampton, Southampton, Southampton, Southampton, Southampt~
## $ country   <fct> United States, United States, United States, England, Norway,~
## $ fare      <dbl> 7.1100, 20.0500, 20.0500, 20.0500, 7.1300, 7.1300, 24.0000, 2~
## $ sibsp     <dbl> 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0~
## $ parch     <dbl> 0, 2, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0~
## $ survived  <fct> no, no, no, yes, yes, yes, no, yes, yes, yes, no, no, no, yes~
```

```
summary(titanic)
```

##	gender	age	class	embarked
##	female: 489	Min. : 0.1667	1st	:317 Belfast : 188
##	male :1690	1st Qu.:22.0000	2nd	:270 Cherbourg : 268
##		Median :29.0000	3rd	:702 Queenstown : 123

```
##           Mean   :30.4075   deck crew       : 66   Southampton:1600
##           3rd Qu.:38.0000   engineering crew:324
##           Max.    :74.0000   restaurant staff: 69
##                               victualling crew:431
##           country      fare      sibsp      parch
## England      :1113   Min.    : 0.000   Min.    :0.0000   Min.    :0.0000
## United States: 262   1st Qu.: 0.000   1st Qu.:0.0000   1st Qu.:0.0000
## Ireland      : 136   Median : 7.151   Median :0.0000   Median :0.0000
## Sweden       : 104   Mean    :19.785   Mean    :0.2997   Mean    :0.2313
## Lebanon      :  71   3rd Qu.:20.111   3rd Qu.:0.0000   3rd Qu.:0.0000
## (Other)      : 413   Max.    :512.061   Max.    :8.0000   Max.    :9.0000
## NA's         :  80
## survived
## no :1470
## yes: 709
##
##
##
##
##
```

2. Define the split between training and testing data

```
set.seed(1234)
training_pct <- 0.75
n <- nrow(titanic)
Z <- sample(1:n, floor(n*training_pct))

dep_train <- titanic[Z,]
dep_test  <- titanic[-Z,]
nrow(dep_train)
```

```
## [1] 1634
```

```
nrow(dep_test)
```

```
## [1] 545
```

3. Run the model on the training data

```
lr_fit_train <- glm(survived ~ age + gender + class,
                    family = binomial, data = dep_train)
summary(lr_fit_train)
```

```
##
## Call:
## glm(formula = survived ~ age + gender + class, family = binomial,
##      data = dep_train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.250152   0.310812  10.457 < 2e-16 ***
## age             -0.031752   0.005782  -5.491 3.99e-08 ***
## gendermale      -2.427474   0.163687 -14.830 < 2e-16 ***
## class2nd        -1.122929   0.243102  -4.619 3.85e-06 ***
## class3rd        -2.212679   0.224567  -9.853 < 2e-16 ***
## classdeck crew   0.877071   0.347642   2.523  0.0116 *
## classengineering crew -0.979696  0.229494  -4.269 1.96e-05 ***
## classrestaurant staff -3.173247  0.637491  -4.978 6.43e-07 ***
## classvictualling crew -1.189151  0.221549  -5.367 7.99e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2050.3  on 1633  degrees of freedom
## Residual deviance: 1583.2  on 1625  degrees of freedom
## AIC: 1601.2
##
## Number of Fisher Scoring iterations: 5
```

4. Get predictions on the test data

```
Probs_test <- predict(object = lr_fit_train,
                      newdata = data.frame(dep_test),
                      type = "response")
summary(Probs_test)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02376 0.15446 0.24123 0.33896 0.49741 0.96033
```

5. Set up the possible thresholds

```
threshold <- seq(0, 1, 0.01)
length(threshold)
```

```
## [1] 101
```

```
head(threshold)
```

```
## [1] 0.00 0.01 0.02 0.03 0.04 0.05
```

6. Test all the possible thresholds

```
TPR <- FPR <- err.rate <- class.rate <- rep(0, length(threshold))

for (i in seq_along(threshold)) {
  Yhat <- rep(NA_character_, nrow(dep_test))
  Yhat <- ifelse(Probs_test >= threshold[[i]], "yes", "no")

  err.rate[i] <- mean(Yhat != dep_test$survived)
  class.rate[i] <- mean(Yhat == dep_test$survived)

  TPR[[i]] <- sum(Yhat == "yes" & dep_test$survived == "yes") /
    sum(dep_test$survived == "yes")
  FPR[[i]] <- sum(Yhat == "yes" & dep_test$survived == "no") /
    sum(dep_test$survived == "no")
}
```

7. Show results

```
err_rate <- data.frame(threshold, err.rate, class.rate, TPR, FPR)
err_rate %>%
  slice_min(err.rate) -> min_err_rate
min_err_rate
```

threshold	err.rate	class.rate	TPR	FPR
0.41	0.1614679	0.8385321	0.6864865	0.0833333

- **Answer:** As it is presented above, the threshold that minimizes the error rate is $p = 0.41$, which allowed for a classification rate of 0.8385 or 83.85% of correct predictions.

b) In class, we deleted fare and embarked from the logistic regression model. Add them back into the model using the just created training and testing data. Describe the results of adding the two variables with respect to the threshold and the optimal prediction error classification rate? Which model would you recommend and why.

```
lr_fit_train_full <- glm(survived ~ age + gender + class + fare + embarked,
                        family = binomial, data = dep_train)
summary(lr_fit_train_full)
```

```
##
## Call:
## glm(formula = survived ~ age + gender + class + fare + embarked,
##      family = binomial, data = dep_train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.052823   0.465255   6.562 5.32e-11 ***
## age             -0.031159   0.005829  -5.345 9.02e-08 ***
## gendermale      -2.433613   0.167242 -14.551 < 2e-16 ***
## class2nd        -1.023607   0.284736  -3.595 0.000324 ***
## class3rd        -2.175780   0.279419  -7.787 6.87e-15 ***
## classdeck crew   1.017594   0.402649   2.527 0.011496 *
## classengineering crew -0.850283  0.297161  -2.861 0.004218 **
## classrestaurant staff -3.048652  0.665617  -4.580 4.65e-06 ***
## classvictualling crew -1.051817  0.297151  -3.540 0.000401 ***
## fare            -0.001207   0.001970  -0.613 0.540013
## embarkedCherbourg  0.592087   0.330575   1.791 0.073280 .
## embarkedQueenstown  0.186129   0.375243   0.496 0.619880
## embarkedSouthampton 0.063594   0.249199   0.255 0.798573
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2050.3  on 1633  degrees of freedom
## Residual deviance: 1577.0  on 1621  degrees of freedom
## AIC: 1603
##
## Number of Fisher Scoring iterations: 5
```

```
Probs_test_full <- predict(object = lr_fit_train_full,
                          newdata = data.frame(dep_test),
```

```

                                type = "response")
summary(Probs_test_full)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02399 0.15907 0.24171 0.34059 0.50561 0.94902

```

```

length(Probs_test_full)

```

```

## [1] 545

```

```

YesPredict <- 1 * (Probs_test_full > 0.41)
table(YesPredict)

```

```

## YesPredict
##    0    1
## 391 154

```

```

tab <- table(dep_test$survived, YesPredict) #confusion matrix.
tab

```

```

##      YesPredict
##           0    1
## no   330   30
## yes   61  124

```

```

TPR <- FPR <- err.rate <- class.rate <- rep(0, length(threshold))

for (i in seq_along(threshold)) {
  Yhat <- rep(NA_character_, nrow(dep_test))
  Yhat <- ifelse(Probs_test_full >= threshold[[i]], "yes", "no")

  err.rate[i] <- mean(Yhat != dep_test$survived)
  class.rate[i] <- mean(Yhat == dep_test$survived)

  TPR[[i]] <- sum(Yhat == "yes" & dep_test$survived == "yes") /
    sum(dep_test$survived == "yes")
  FPR[[i]] <- sum(Yhat == "yes" & dep_test$survived == "no") /
    sum(dep_test$survived == "no")
}

```

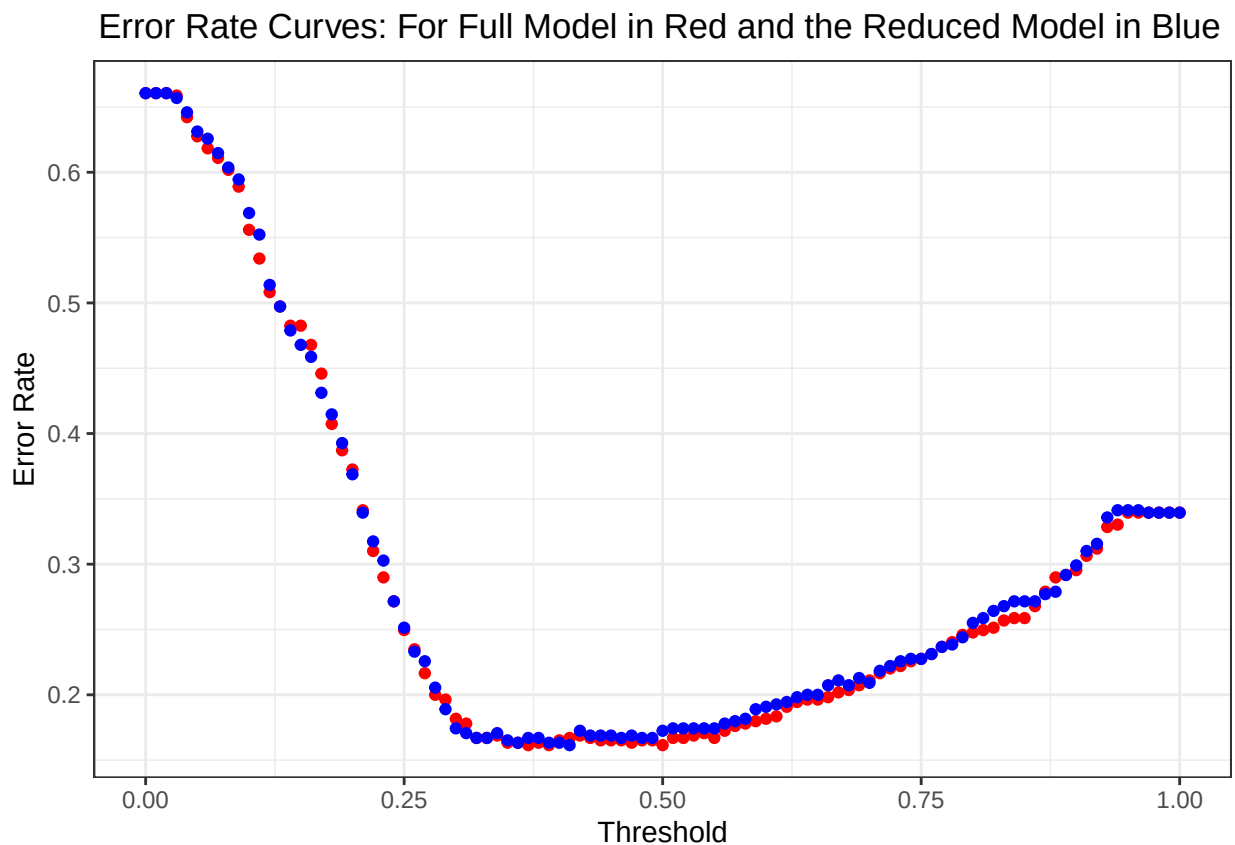
```
err_rate_F <- data.frame(threshold, err.rate, class.rate, TPR, FPR)
err_rate_F %>%
  slice_min(err.rate) -> min_err_rate_F
min_err_rate_F
```

threshold	err.rate	class.rate	TPR	FPR
0.37	0.1614679	0.8385321	0.6972973	0.08888889
0.39	0.1614679	0.8385321	0.6864865	0.08333333
0.50	0.1614679	0.8385321	0.6324324	0.05555556

- **Answer:** The variables **fare** and **embarked** are not statistically significant to the model at 0.05 level and, consequently, they should not influence the survival rate. Considering such variables in the model changed the threshold from 0.41 to three different thresholds 0.37, .039 and 0.50, but allowing for the same error rate and classification rate of 16.15% and 83.85%, respectively. In this case, it's confusing to choose the best threshold for the analysis. The first model, without **fare** and **embarked**, is preferred since these variables didn't add anything to the model and caused confusion instead.

c) Generate a single plot showing the error rate curves for both models. Use color to differentiate the curves and label (in the title or by other means). Interpret the plot to answer if either model is vastly better than the other.

```
ggplot(err_rate_F, aes(threshold, err.rate)) +  
  geom_point(color = "red", show.legend = TRUE) +  
  geom_point(data = err_rate, mapping = aes(threshold, err.rate), color = "blue") +  
  theme_bw() +  
  xlab("Threshold") +  
  ylab("Error Rate") +  
  ggtitle(label = "Error Rate Curves: For Full Model in Red and the Reduced Model in Blue") +  
  theme(plot.title = element_text(hjust = 0.3))
```



- **Answer:** Based on the plot, both models performed pretty much the same way. However, the model without `fare` and `embarked` would be preferred because it provides a clear threshold for testing and more reliable results.

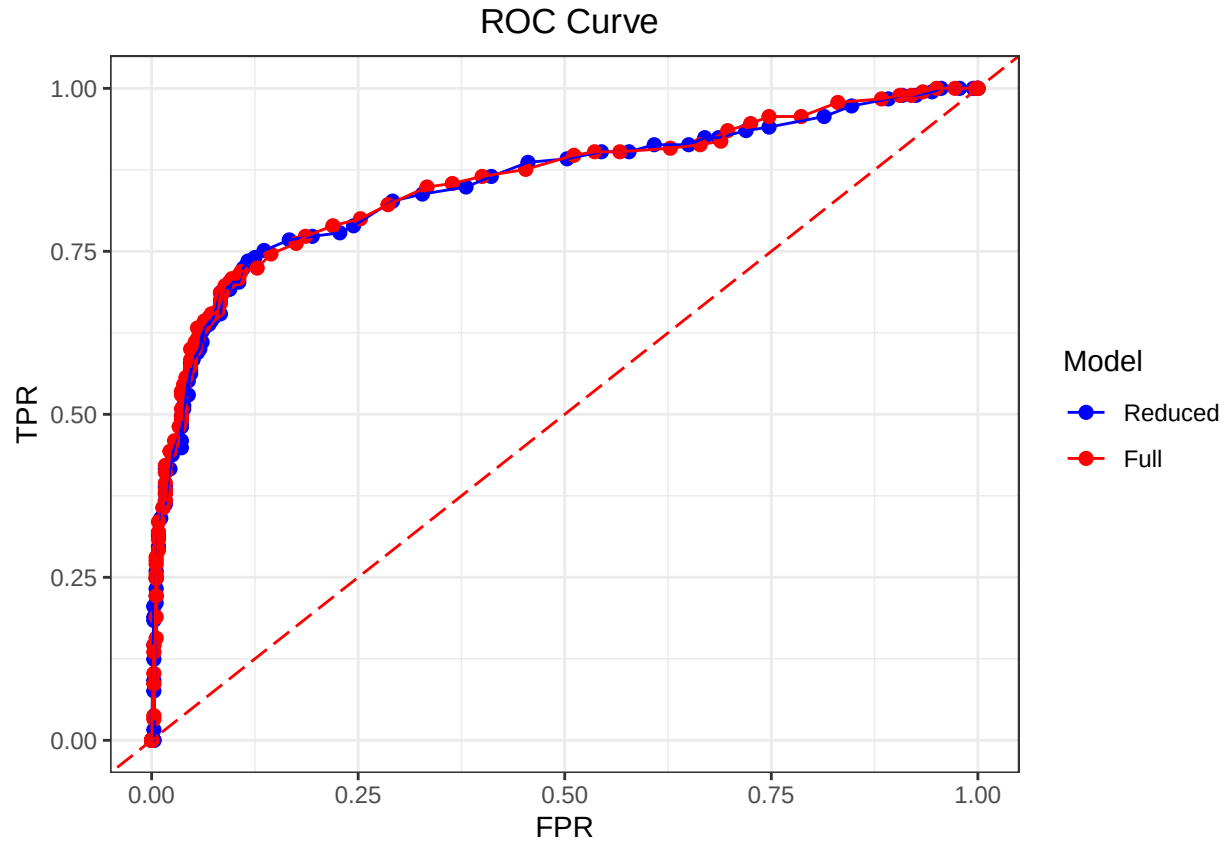
d) STAT 627 Only: Create a tibble/data frame with three columns: the true positive rates for the two models, the false positive rate for the two models and a third column of text identifying the two models.

- Generate a single plot with the ROC curves for the two models. Interpret the plot to answer if either model is much better than the other.

```
combined_data <- bind_rows(err_rate, err_rate_F, .id = "Model")
glimpse(combined_data)
```

```
## Rows: 202
## Columns: 6
## $ Model      <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",~
## $ threshold  <dbl> 0.00, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09,~
## $ err.rate    <dbl> 0.6605505, 0.6605505, 0.6605505, 0.6568807, 0.6458716, 0.63~
## $ class.rate  <dbl> 0.3394495, 0.3394495, 0.3394495, 0.3431193, 0.3541284, 0.36~
## $ TPR         <dbl> 1.0000000, 1.0000000, 1.0000000, 1.0000000, 1.0000000, 1.00~
## $ FPR         <dbl> 1.0000000, 1.0000000, 1.0000000, 0.9944444, 0.9777778, 0.95~
```

```
ggplot(combined_data, aes(x = FPR, y = TPR, color = Model)) +
  geom_point(size = 2) +
  geom_line() +
  geom_abline(color = "red", lty = 5) +
  ylim(0, 1) +
  xlim(0, 1) +
  theme_bw() +
  scale_color_manual(
    values = c("1" = "blue", "2" = "red"),
    labels = c("1" = "Reduced", "2" = "Full")) +
  xlab("FPR") +
  ylab("TPR") +
  ggtitle(label = " ROC Curve") +
  theme(plot.title = element_text(hjust = 0.5))
```



Answer:

- The curve rises above the red diagonal line, indicating the model performs better than random guessing.
- The curve is somewhat steep or close to the top-left corner, so the model seems to be good in predicting survival rates.
- The curve shows a steep increase in TPR as FPR increases and an indication for the best threshold at the "elbow" or "knee".
- In the lower FPR range (left corner), the curve rises quickly, suggesting the model is better at identifying positives early at lower false positive rates.

4 Stock Market Prediction

Page 193, chap. 4, #13 (selected questions).

We want to predict the behavior of the stock market in the following week. The `Weekly` data set, (from the `{ISLR2}` package), contains 1,089 observations with the following 9 variables.

- **Year**: The year that the observation was recorded
- **Lag1**: Percentage return for previous week
- **Lag2**: Percentage return for 2 weeks previous
- **Lag3**: Percentage return for 3 weeks previous
- **Lag4**: Percentage return for 4 weeks previous
- **Lag5**: Percentage return for 5 weeks previous
- **Volume**: Volume of shares traded (average number of daily shares traded in billions)
- **Today**: Percentage return for this week
- **Direction**: A factor with levels `Down` and `Up` indicating whether the market had a positive or negative return on a given week

```
glimpse(Weekly)
```

```
## Rows: 1,089
## Columns: 9
## $ Year      <dbl> 1990, 1990, 1990, 1990, 1990, 1990, 1990, 1990, 1990, 1990, ~
## $ Lag1      <dbl> 0.816, -0.270, -2.576, 3.514, 0.712, 1.178, -1.372, 0.807, 0~
## $ Lag2      <dbl> 1.572, 0.816, -0.270, -2.576, 3.514, 0.712, 1.178, -1.372, 0~
## $ Lag3      <dbl> -3.936, 1.572, 0.816, -0.270, -2.576, 3.514, 0.712, 1.178, ~~
## $ Lag4      <dbl> -0.229, -3.936, 1.572, 0.816, -0.270, -2.576, 3.514, 0.712, ~
## $ Lag5      <dbl> -3.484, -0.229, -3.936, 1.572, 0.816, -0.270, -2.576, 3.514,~
## $ Volume     <dbl> 0.1549760, 0.1485740, 0.1598375, 0.1616300, 0.1537280, 0.154~
## $ Today      <dbl> -0.270, -2.576, 3.514, 0.712, 1.178, -1.372, 0.807, 0.041, 1~
## $ Direction <fct> Down, Down, Up, Up, Up, Down, Up, Up, Up, Down, Down, Up, Up~
```

```
summary(Weekly)
```

##	Year	Lag1	Lag2	Lag3
##	Min. :1990	Min. : -18.1950	Min. : -18.1950	Min. : -18.1950
##	1st Qu.:1995	1st Qu.: -1.1540	1st Qu.: -1.1540	1st Qu.: -1.1580
##	Median :2000	Median : 0.2410	Median : 0.2410	Median : 0.2410
##	Mean :2000	Mean : 0.1506	Mean : 0.1511	Mean : 0.1472
##	3rd Qu.:2005	3rd Qu.: 1.4050	3rd Qu.: 1.4090	3rd Qu.: 1.4090
##	Max. :2010	Max. : 12.0260	Max. : 12.0260	Max. : 12.0260
##	Lag4	Lag5	Volume	Today

##	Min.	:-18.1950	Min.	:-18.1950	Min.	:0.08747	Min.	:-18.1950
##	1st Qu.:	-1.1580	1st Qu.:	-1.1660	1st Qu.:	0.33202	1st Qu.:	-1.1540
##	Median :	0.2380	Median :	0.2340	Median :	1.00268	Median :	0.2410
##	Mean :	0.1458	Mean :	0.1399	Mean :	1.57462	Mean :	0.1499
##	3rd Qu.:	1.4090	3rd Qu.:	1.4050	3rd Qu.:	2.05373	3rd Qu.:	1.4050
##	Max.	: 12.0260	Max.	: 12.0260	Max.	:9.32821	Max.	: 12.0260
##	Direction							
##	Down:484							
##	Up :605							
##								
##								
##								
##								

a) Use all the data as training data and perform a logistic regression with Direction as the response and the five lag variables plus Volume as predictors. Do any of the predictors appear to be statistically significant? If so, which ones?

```
smw_logit <- glm(Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
                 family = binomial,
                 data = Weekly)
summary(smw_logit)
```

```
##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##      Volume, family = binomial, data = Weekly)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.26686    0.08593   3.106  0.0019 **
## Lag1        -0.04127    0.02641  -1.563  0.1181
## Lag2         0.05844    0.02686   2.175  0.0296 *
## Lag3        -0.01606    0.02666  -0.602  0.5469
## Lag4        -0.02779    0.02646  -1.050  0.2937
## Lag5        -0.01447    0.02638  -0.549  0.5833
## Volume      -0.02274    0.03690  -0.616  0.5377
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1496.2  on 1088  degrees of freedom
## Residual deviance: 1486.4  on 1082  degrees of freedom
## AIC: 1500.4
##
## Number of Fisher Scoring iterations: 4
```

- **Answer:** Based on the p-values from the output above, only the Lag2 predictor is statistical significant at 0.05 level, with p-value = 0.0296.

b) Compute the confusion matrix and overall fraction of *correct* predictions. Explain what the confusion matrix is telling you about the types of mistakes made by the logistic regression model.

```
Probs_smw_logit <- predict(object = smw_logit, type = "response")
summary(Probs_smw_logit)
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.2864  0.5277  0.5574  0.5556  0.5819  0.7837
```

```
glm_pred_smw <- rep("Down", nrow(Weekly))
glm_pred_smw[Probs_smw_logit > 0.5] = "Up"
```

```
tab <- table(Weekly$Direction, glm_pred_smw) # confusion matrix
tab
```

```
##      glm_pred_smw
##      Down  Up
## Down   54 430
## Up    48 557
```

```
mean(glm_pred_smw == Weekly$Direction) # fractions of correct predictions.
```

```
## [1] 0.5610652
```

```
mean(glm_pred_smw != Weekly$Direction)
```

```
## [1] 0.4389348
```

- **Answer:** The considered model correctly predicted the direction of the stock market 56.10 % of the time, while getting it wrong on 43.90% of the time. Therefore, it looks like this logistic regression model is working better than random guessing alone. However, only the predictor `Lag2` seems to have an association with the response variable `Direction` and all the other variables can be dropped from the model without any cost.

c) Fit a logistic regression model with Lag2 as the only predictor using as training data the period from 1990 to 2007 (inclusive). Compute the confusion matrix and the overall fraction of correct predictions for the held out test data (that is the data from 2008, 2009, and 2010). How does it compare to before? What does that suggest?

```
Weekly %>%
  filter (Year >= 1990 & Weekly$Year <= 2007) -> weekly_train
nrow(weekly_train)
```

```
## [1] 933
```

```
Weekly %>%
  filter (Year > 2007) -> weekly_test
nrow(weekly_test)
```

```
## [1] 156
```

```
smw_logit_red <- glm(Direction ~ Lag2,
                     family = binomial,
                     data = weekly_train)
summary(smw_logit_red)
```

```
##
## Call:
## glm(formula = Direction ~ Lag2, family = binomial, data = weekly_train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.22658    0.06621   3.422 0.000621 ***
## Lag2         0.04716    0.03230   1.460 0.144293
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1280.6  on 932  degrees of freedom
## Residual deviance: 1278.5  on 931  degrees of freedom
## AIC: 1282.5
##
## Number of Fisher Scoring iterations: 4
```

```
Probs_test_Red <- predict(object = smw_logit_red,
                          newdata = data.frame(weekly_test),
                          type = "response")
summary(Probs_test_Red)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3472  0.5332  0.5575  0.5555  0.5803  0.6886
```

```
glm_pred_test <- rep("Down", nrow(weekly_test))
glm_pred_test[Probs_test_Red > 0.5] = "Up"

tab <- table(weekly_test$Direction, glm_pred_test) # confusion matrix
tab
```

```
##      glm_pred_test
##      Down Up
## Down      7 65
## Up       5 79
```

```
mean(glm_pred_test == weekly_test$Direction) # fractions of correct predictions.
```

```
## [1] 0.5512821
```

```
mean(glm_pred_test != weekly_test$Direction)
```

```
## [1] 0.4487179
```

- **Answer:** The correct prediction rate is actually lower than before, about 0.5513 or 53.13%. Hence, keeping only `Lag2` in the model, while dropping the years 2008, 2009 and 2010, did not improve the model when compared with before. Also, the predictor `Lag2` is no longer statistically significant at 0.05 level and the model is not good in predicting survival rate with it's only predictor not showing an association with the response variable `Survival`. I would suggest a random selection for the observations of the training and testing datasets and running the model again.

d) Plot an ROC curve for the logistic regression on the test data from (c), using different probability thresholds and add an abline for $FPR = TPR$ (colored red). Interpret the plot in terms of how useful the model might be.

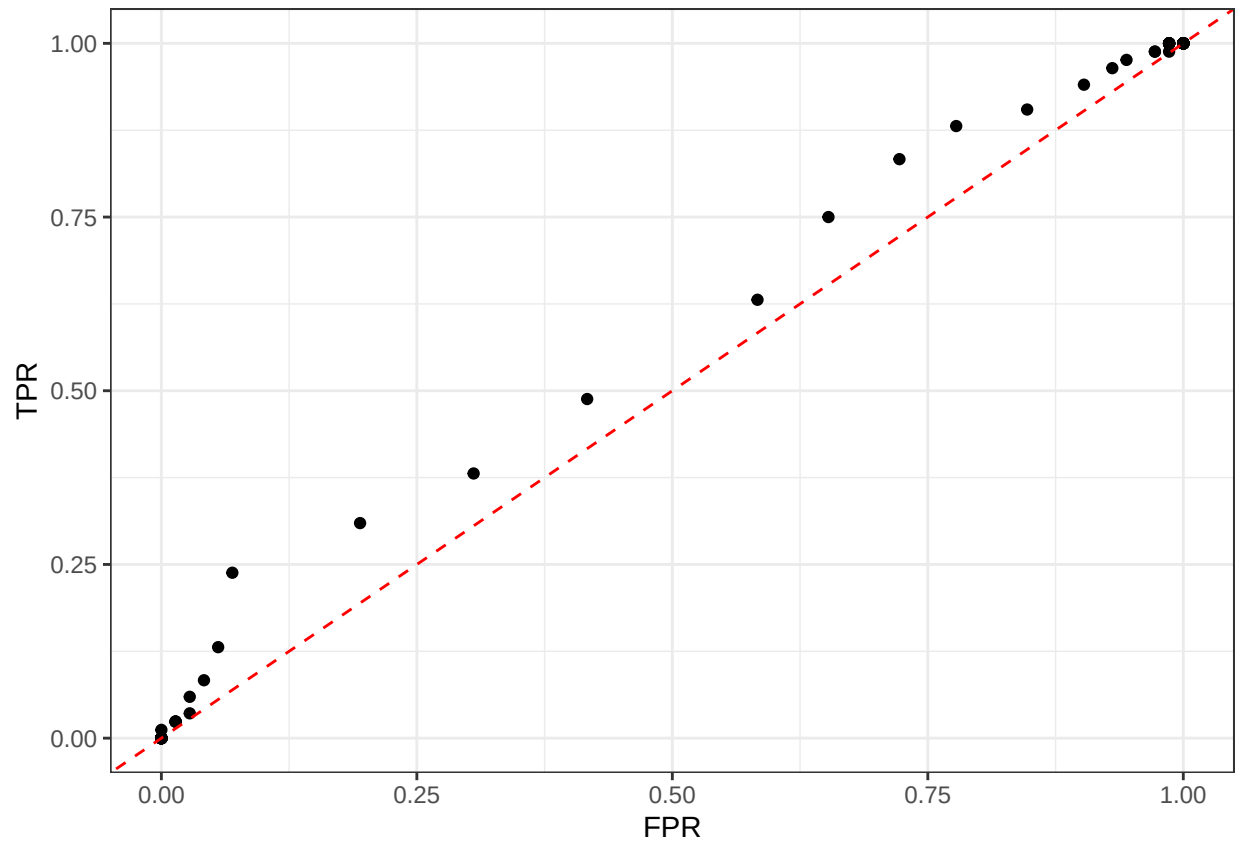
```
threshold <- seq(0, 1, 0.01)

TPR <- FPR <- rep(0, length(threshold))

for (i in seq_along(threshold)) {
  Yhat <- rep(NA_character_, nrow(weekly_test))
  Yhat <- ifelse(Probs_test_Red >= threshold[[i]], "Up", "Down")

  TPR[[i]] <- sum(Yhat == "Up" & weekly_test$Direction == "Up") /
    sum(weekly_test$Direction == "Up")
  FPR[[i]] <- sum(Yhat == "Up" & weekly_test$Direction == "Down") /
    sum(weekly_test$Direction == "Down")
}

ggplot(tibble(TPR, FPR), aes(FPR, TPR)) +
  geom_point() +
  geom_abline(color = "red", lty = 2) +
  theme_bw()
```



- **Answer:** From the plot above, the ROC curve is not good and the model isn't reliable at all in predicting survival rate.